RESEARCH OF EVGENY FEIGIN

Evgeny Feigin was my PhD student during the period 2002-2005. He defended his thesis "Demazure modules and Schubert varieties for the Lie algebra $\widehat{\mathfrak{sl}}_2$" in 2005.

Mathematical interests of E.Feigin are concentrated around Lie theory and infinite-dimensional algebras, such as affine Kac-Moody algebras, current algebras and vertex operator algebras. He studies representation theory of these algebras and applications to algebraic geometry and mathematical physics. I am providing some details below.

One of the main objects of research of E.Feigin are integrable representations of affine Kac-Moody algebras. These representations are infinite-dimensional analogues of simple representations of finite-dimensional Lie algebras. They play an important role in lattice models and conformal field theories. In particular, an important problem is to compute characters of a representation itself as well as of its certain subspaces. E.Feigin studied classes of finite-dimensional subspaces (Demazure modules) and of infinite-dimensional subspaces (principal subspaces). He obtained various character formulas for these spaces. He used his formulas to compute Littlewood-Richardson coefficients for general affine Kac-Moody algebra and gave fermionic and bosonic expressions for corresponding characters. E.Feigin also obtained several formulas for Kostka polynomials using principal subspaces in integrable modules. He also found a remarkable connection between bosonic formulas and Euler characteristics of certain complexes, based on the BGG resolutions.

In his recent papers Evgeny Feigin studied certain structures coming from the double-loop (toroidal) generalizations of Kac-Moody theory and a connection between representation theory of affine algebras and quantum integrable systems. In particular, he studied the Poincaré-Birkhoff-Witt filtration on vacuum level one modules, which turned out to be important in the study of $(1,3)$ vertex operator in minimal conformal field theories. This filtration is also used for a description of graded version of tensor products of integrable representation of affine algebras, thus providing important examples of representations of toroidal algebras. Another subject of his recent studies are quantum finite-difference Toda equations. In particular, new formulas for the eigenfunctions of the corresponding Hamiltonians were obtained.

I recommend Evgeny Feigin as a participant of the summer school "Structures in Lie representation theory".

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